

PLASMA SPRAY PROCESSORS Works Instruction	QF/QMS/02/39	
	Rev. 00	
	Date : 25.04.2024	

Operations/Work/Job Activity covered by this assessment: Macro Hardness Tester (Rockwell)

Required Personal Protective Equipment (PPE) :



Safety glasses must be worn at times.



Long and loose hair must be contained.



Appropriate footwear with substantial uppers must be worn.



Close fitting/protective clothing to cover arms and legs must be worn.



Respiratory protection devices required.



Oil free leather gloves and spats must be worn.



Hearing protection must be used when using this machine.

1. Identify the hardness scale required for the sample under testing.(HRA,HRB,HRC,HR15N & HR15Y)(A)Fig.1
2. Turn the knob to Rockwell position for scales HRA,HRB & HRC and Superficial position for scales HR15N & HR15Y (B) (Fig.2 a,b)
3. Select the load based on the selected scale by rotating the knob.(C) (Table 1)
4. Mount the suitable indenter based on selected scale.(D)Refer table 1.
5. Clean the test-block with tissue paper and place it on the special anvil or work table of the machine.(E)
6. Ensure the loading lever is pulled forward as shown in fig (Fig.3)
7. Rotate the capston wheel slowly to raise the test specimen till it comes in contact with the indenter tip.(F)
8. Ensure the corresponding scale is set at zero.(A)
9. Push back the loading lever as shown in fig.(Fig.4)
10. After time lapse of 5 seconds record the reading shown on the scale.
11. Repeat the steps shown in Sr.no 5 to 8 three times and record the readings.
12. From the readings obtained from sr.no 11, record the deviation from the standard value.
13. Now the specimen is kept on the anvil and steps 5 to 8 are repeated.
14. After recording the values, pull forward the loading lever to the unloading position.(Fig.3)



Assessor	Prasanna Wadkar	Signature:		Date:	
Approved:	Haarsh Doshi	Signature:		Date:	

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(Fig.2 a)



(Fig.2 b)



(Fig.3)



(Fig.4)

SCALE	LOAD	INDENTOR	DIAL SCALE
HRA	60 Kg	DIAMOND	BLACK
HRB	100 Kg	1/16" BALL	RED
HRC	150 Kg	DIAMOND	BLACK
HR15Y	15 Kg	½" BALL	N/T
HR15N	15 Kg	DIAMOND	N/T

(Table.1)

Assessor	Prasanna Wadkar	Signature:		Date:	
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